

BaaliMitra: An AI-Driven Multimodal Agricultural Intelligence System for Precision Agriculture in Nepal

Anupam Chhetri

*anupamdevx@gmail.com

Computer and Artificial Intelligence

1. INTRODUCTION

Agriculture in Nepal is an important activity, employing around 60% of the population and contributing about one-third of the GDP. It is one of the major sectors that influences a country's economic growth. But Farmers often lack access to monitor their soil data leaving them uncertain about what type of crop to sow, what fertilizer to use to maximize their yield. Even after knowing the proper crops to sow, farmers are unable to properly monitor their crop health in time leading to inefficient productivity (Bhansali, Shah et al. 2022).

With many advancements in Machine learning and Deep learning field alongside IoT, we can effectively implement an ecosystem to help farmers of Nepal to successfully maximize their yield and turn farming into a profitable business. Remote sensing and multispectral techniques further enhance crop health monitoring (Rodríguez, Lizarazo et al. 2021).

Thus, BaaliMitra uses supervised learning models on leaf and soil data, with multispectral images as an optional extension and a voice-based interface to address different problems farmer face.

2. SYSTEM ARCHITECTURE & DEVELOPMENT

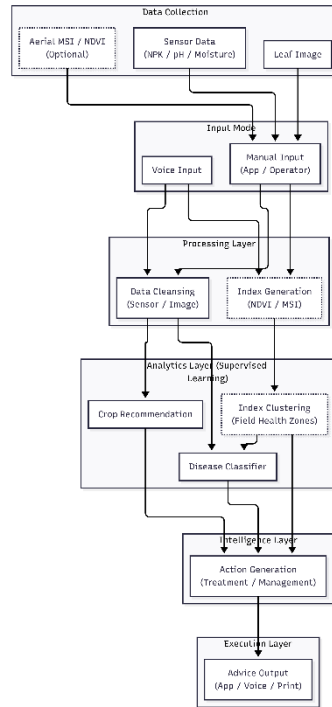


Figure 1: Working pipeline

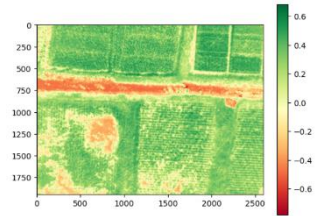


Figure 2: Example vegetation index (NDVI) (Fonseka, Halloluwa et al. 2024)

2.1 PROCESSING PIPELINE

As shown in Figure 1, our system is mainly dependent on two data streams. Leaf images and soil NPK,pH and moisture values.

UAV-obtained multispectral images can be treated as optional extensions which are converted into vegetation indexes such as NDVI and NDWI, providing spatial indicators of vegetation health and moisture levels (Paredes, González et al. 2017). Leaf images obtained from UGV, or smartphones are standardized to ensure compatibility with the model. Sensor data are cleansed to prevent extreme variations, producing false analysis whilst environmental data is synchronized.

2.2 ANALYTICAL AND DECISION FRAMEWORK

Analytical layer integrates three supervised-learning components. Firstly, an ML model used to provide crop recommendations using NPK values and regional parameters. Secondly, vegetation index-based clustering may be applied to aerial MSI images which further divide fields into health zones, assisting and enhancing crop health monitoring. Lastly, transfer learning-based CNN model, used for crop disease detection using leaf. The models mentioned are not treated independently. Instead, a lightweight fusion mechanism is used which suggests farmer treatment advice. This improves real-world adaptability across both traditional and NewGen farmers.

3. RESULTS AND DISCUSSION

Table: Comparing disease detector model accuracy and technique used

S.N.	Crop Name	TL-Models	Accuracy
1	Potato	MobileNetV2	99.52%
2	Rice	DenseNet121	95.20%
3	Tomato	ResNet50V2	98.31%

Among the multi-model approach used in our system; the given data illustrates the accuracy of disease prediction model. Comprising leaf-vs-non leaf detector, species classifier and disease-prediction, each of the models obtained over 90% accuracy using publicly available PlantVillage dataset, enhanced by augmenting it. The crop recommendation model also showed significant results and potential.

Different vegetation indexes generated from multispectral images obtained from showed distinct field health zones whilst captured images supports disease detection through vision pipeline. UAV and UGV provide enhanced accuracy optimizing the accuracy of our overall model.

4. CONCLUSION

Overall, BaaliMitra illustrates use of Computer and AI can help farmers in crop health monitoring and decision-making. The system shows strong performance and highlights its potential in precision agriculture in Nepal. Furthermore, expanding research on real-field deployment, datasets and integrating real time sensors and robotic feedback promotes its impact in modern AI agriculture.

5. ACKNOWLEDGEMENT

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6. REFERENCES

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